1.5 Infinite Limits; Limits at Infinity; Asymptotes

Name _____ Period ____ Date _____

AP Calculus 1.5 · Student sheet

I can…

- find infinite limits and vertical asymptotes.
- find limits at infinity of rational functions.
- find the horizontal and vertical asymptotes of a rational function.

Vocabulary · fill in the missing word

Infinite limit — $f(x) \to \infty$ (or $-\infty$) as $x \to c$: $f(x)$ _____ without bound

Vertical asymptote — the line $x =$ _____

Limit at infinity — _____

Horizontal asymptote — the line $y =$ _____

Part 1 · Big and small

Fill in each table.

x	±0.1	±0.01	±0.001
$1/x^2$			

x	10	100	1000	10,000
$(2x + 1)/x$				

Part 2 · Notes and examples (fill in each note, then work the example)

Vertical asymptote: if $f(x) \to \pm\infty$ as $x \to c^-$ or c^+, the line $x = c$ is a _____ asymptote.

Example 1

Find the limit of $\dfrac{5}{x^2 - 4}$ as $x \to 2^+$ and as $x \to 2^-$.

Example 2

Find the limit as $x \to -1$ of $\dfrac{x + 2}{(x + 1)^2}$.

Limits at infinity: $1/x^p \to$ _____ as $x \to \pm\infty$, for $p > 0$.

Rational functions at infinity: divide the numerator and the denominator by the highest power of x in the _____.

Example 3

Find each limit as $x \to \infty$.

a. $\dfrac{2x + 4}{5x}$

b. $\dfrac{x^3 + x^2 + 2x - 1}{x^3 + x + 1}$

Example 4

Find each limit as $x \to \infty$.

a. $\dfrac{2x^2 - 5x + 2}{5x^3 + 1}$

b. $\dfrac{x^2 + 1}{x - 1}$

Horizontal asymptote: $y = L$ is a horizontal asymptote if $f(x) \to L$ as $x \to \infty$ or as $x \to$ _____.

Example 5

Find the asymptotes of $R(x) = \dfrac{x^2 + 3x - 10}{2x^2 - 7x + 6}$.

Part 3 · Partner practice

Partner 1

Find the limit as $x \to -\infty$ of $\dfrac{x^3 + 1}{x^2}$.

Partner 2

(AP® Practice) The limit as $x \to \infty$ of $\dfrac{3 - 2x^2}{5x^2 + x}$ is

(A) $-\dfrac{2}{5}$ (B) 0 (C) $\dfrac{3}{5}$ (D) ∞